

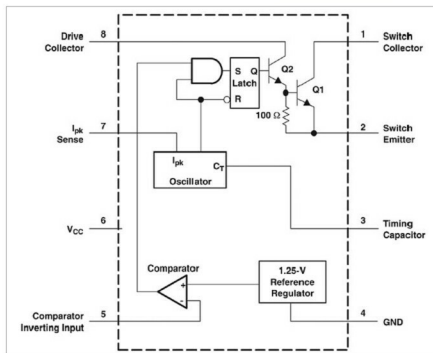


Fig. 4: Block diagram of MC34063

As shown in the block diagram, the output switch is an npn Darlington transistor. The collector is tied to pin 1, and the emitter is tied to pin 2. This allows the designer to use MC34063 in buck, boost or inverter configurations. The maximum collector-emitter saturation voltage at 1.5A (peak) is 1.3V, and the maximum peak current of the output switch is 1.5A. For higher peak output current, an external transistor can be used. The oscillating pulses drive the internal transistors, which may be used to provide boost/buck conversion or to drive an external power transistor of higher rating to get higher power rating.

Some circuit designs, mainly step-up and voltage-inverting, require ton/(ton + toff) ratio greater than 0.857. This can be obtained by adding a ratio extender circuit, which uses germanium diode and is temperature-sensitive. A negative-temperature-coefficient timing capacitor will help reduce this sensitivity. In Fig. 3, the extender circuit consists of transistor T2 (BC557), germanium diode D2 (1N34A) and timing capacitor C3. Here, T2 is not driving anything but a discharging and charging switch for capacitor C3 powered by pin 3 of the IC. Current limiting must be used on all step-up and voltage-inverting designs using the ratio extender circuit.

This allows the inductor time to reset between cycles of over-current during initial power-up of the switcher. When the output filter capacitor reaches its nominal voltage, the voltage feedback loop controls regulation.

In the main circuit, a wire is connected between the junction of resistors R1 and R2 and capacitors C1 and C2 for charge balancing in both output capacitors. Only resistor R3 connected to pin 5 of MC34063 forms the voltage divider.

Software

We have designed a utility software to find out component values quicker for rapid prototyping of a power supply based on MC34063. The software program is written using HTML and JavaScript, and can be incorporated in a system having PHP installed. It runs under PHP development environment. HTML is a front-end software, whereas PHP is a back-end software. JavaScript file checks for blank fields in HTML form. PHP is incorporated in a web server. So multiple user clients within a network, such as in a lab or college, can use this software simultaneously. The program is developed using NetBeans IDE for PHP.

Software installation.

1. Download WampServer (for

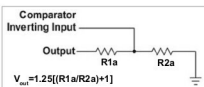


Fig. 5: External resistors

PARTS LIST

Semiconductors:

- MC34063 DC-DC converter IC
- IC1
- FR107 rectifier diode
- D1
- 1N34A germanium diode
- D2
- BUT11A npn power transistor or equivalent
- T1
- BC557 pnp transistor
- T2

Resistors (all 1/4-watt, $\pm 1\%$ metal film, unless stated otherwise):

- R1, R2 - 1.2-mega-ohm
- R3 - 6-kilo-ohm
- R4 - 100-ohm
- R5 - 12-ohm
- R6 - 1.5-ohm, 2-watt

Capacitors:

- C1, C2 - 33 μ F, 500V electrolytic
- C3 - 4.4nF ceramic disk
- C4 - 4700 μ F, 50V electrolytic

Miscellaneous:

- L1 - 6.8mH inductor, drum-core, high-current type
- CON1, CON2 - 2-pin terminal connector
- 8-pin IC socket
- DC jack
- Heat-sink for power transistor BUT11A

localhost development) from www.wampserver.com/en/ and NetBeans IDE from <https://netbeans.org/features/php/>. Install these in your Windows PC. Install appropriate Visual C++ extension (here VC++ 2012) before installing WampServer to get all prerequisite dll files for smooth running of the IDE and Apache server environment.

2. WampServer runs in the background with the option in online mode. Make sure WampServer icon on the taskbar turns green.

3. Create a folder, say, HighVoltage, under C:\wamp\www folder. Copy HVBoostCalculator.html, HVDesign.js and HVCircuit.jpg image files into the project folder.

4. Create a new PHP project in NetBeans. Select 'PHP Application' option and click 'Next'. A project folder will create automatically. Note that your HighVoltage folder is under this project folder.

5. Under 'Run Configuration' window, select 'local server' option